

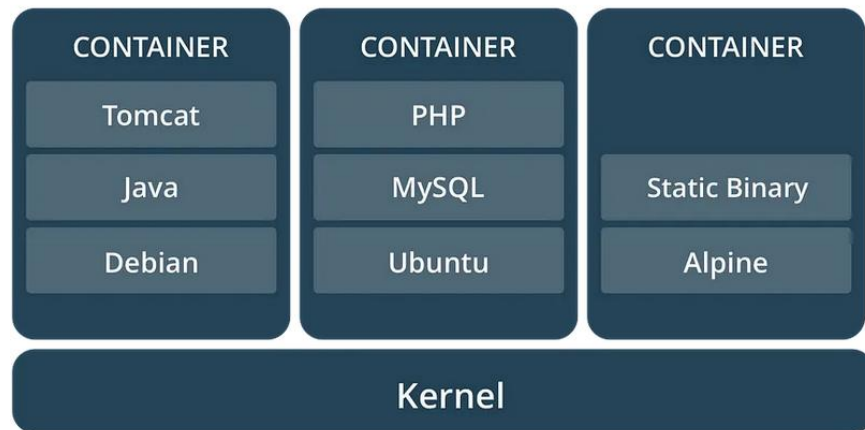
Winget Docker webgoat Zenmap

HungYu Chien

20250916/20240920

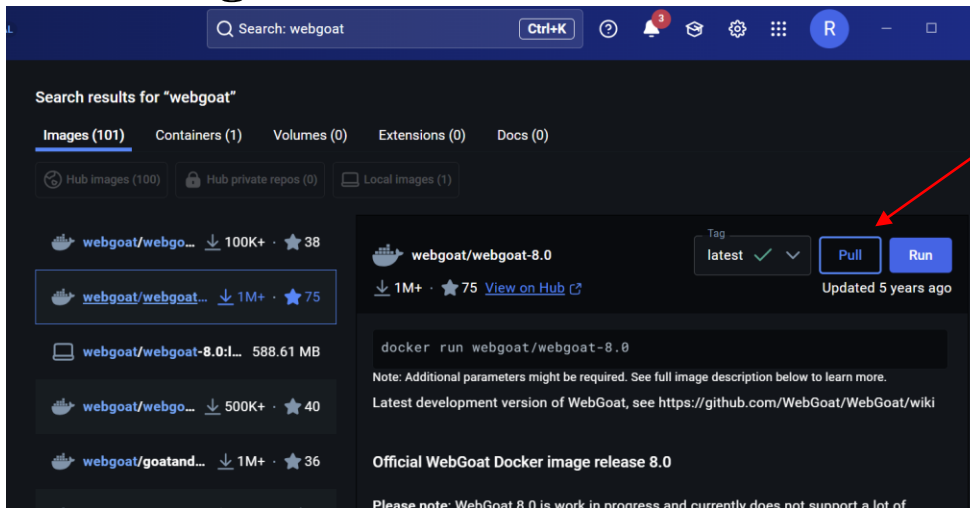
Docker 是什麼

- 輕量級的虛擬化技術
- 底層使用 cgroup、chroot、namespace 實作
- 把你的程式跟環境包成一個 image
- 每個 container 都是一個獨立的環境
- image 指的是已經打包好的環境，但是還沒有跑起來，就是一堆檔案而已，而 container 則是把 image 跑起來之後產生的 instance

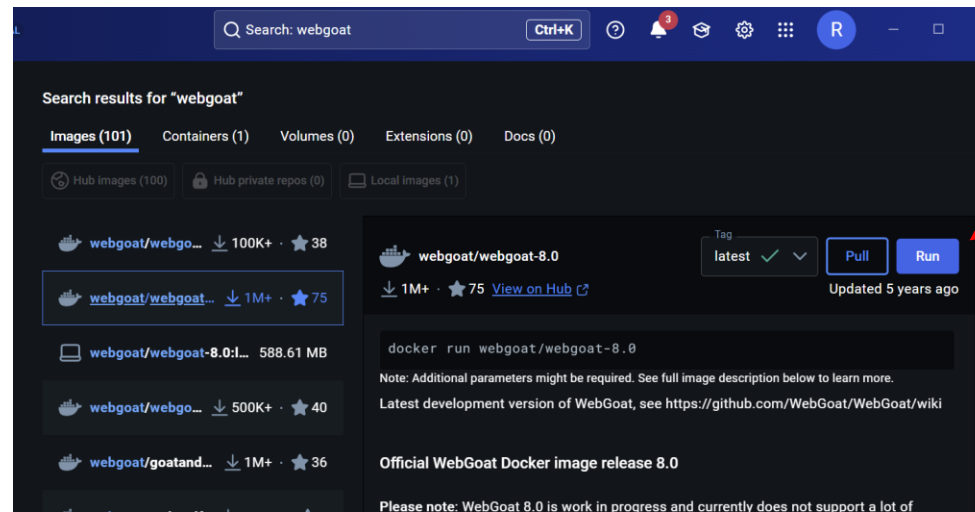


2025Winget install Docker.DockerDesktop

1. Winget install Docker.DockerDesktop → 以 google cloud mail hychien@mail.ncnu.edu.tw
2. Search webgoat → 找到那個 webgoat/webgoat-8.0 那版
3. Pull image

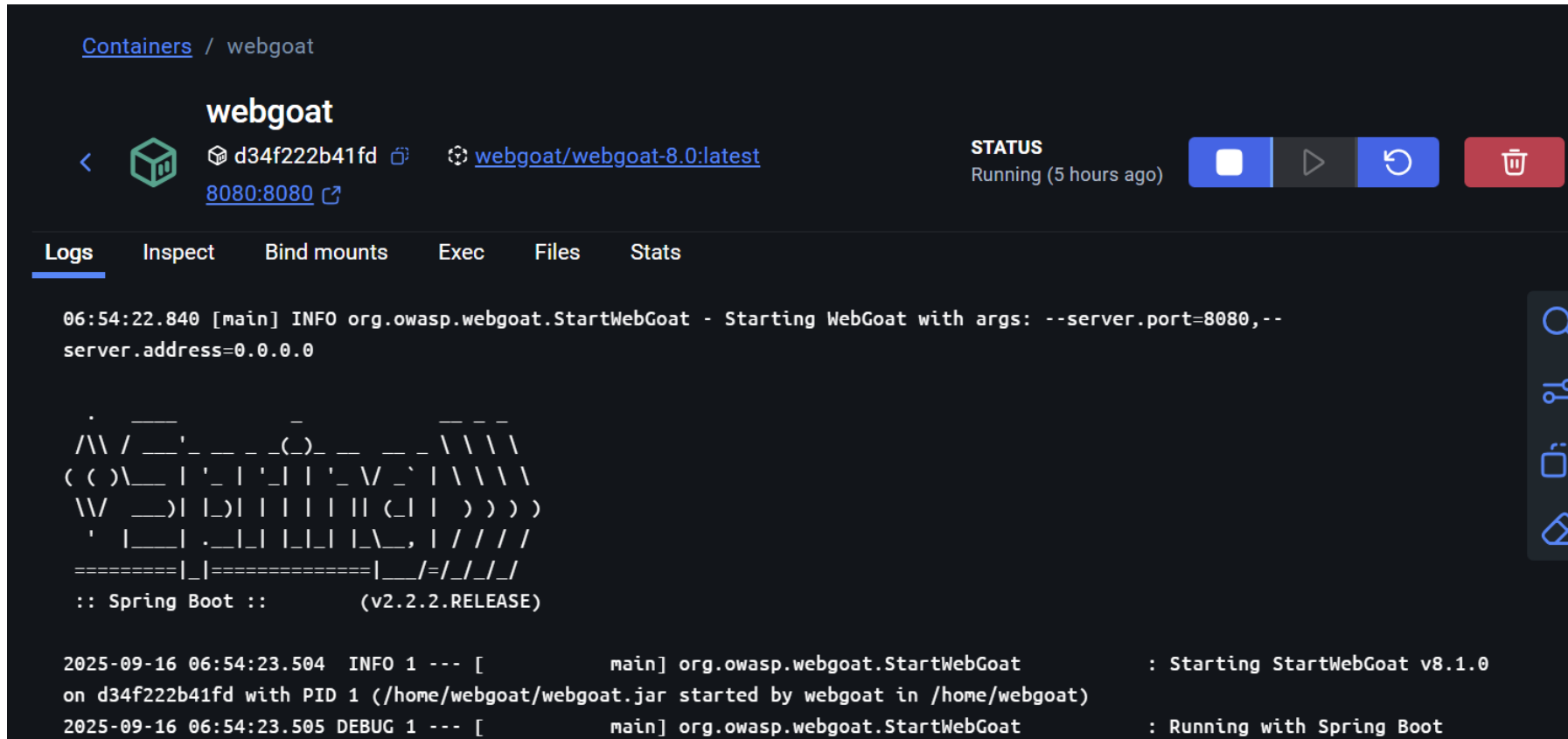


4. Run container



2025 DockerDesktop Webgoat

4. Run container



5. 2025 年起預設啟動 沒有做 docker --> windows map; windows 無法存取 ➡ 參考下一頁

開啟 docker webgoat windows mapping

-2025

查詢 Chat 從 windows access webgoat

ChatGPT 5 ▾

我在windows 作業系統下 透過 winget 安裝 docker.desktop,然後在 docker 下安裝了 webgoat-8.0 container 並啟動, 我接下來可如何用瀏覽器連到 webgoat container

Docker → windows port mapping 開啟

檢查目前 container 狀態

打開 PowerShell 或 CMD，輸入：

```
powershell
```

複製程式碼

```
docker ps
```

你會看到類似：

```
bash
```

複製程式碼

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
abcd1234efgh	webgoat/webgoat-8.0	...	...	...	0.0.0.0:8080->8080/tcp	webgoat

重點是 **PORTS** 欄位，像上面例子表示：

- 容器內的 **8080** port 已經映射到主機 (Windows) 的 **8080** port。

開啟 docker webgoat windows mapping

-2025

瀏覽器存取

如果容器啟動時有做 port mapping，例如：

```
powershell  
  
docker run -d -p 8080:8080 webgoat/webgoat-8.0
```

那麼你就可以直接打開瀏覽器，輸入：

```
arduino  
  
http://localhost:8080/WebGoat
```

若沒開啟 透過指令開啟

有些情況 `docker run` 時沒加 `-p`，容器內的 port 不會自動開放給 Windows。
這時需要重新 run 容器，加入 port mapping，例如：

```
powershell  
  
docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0
```

然後再用瀏覽器連：

```
arduino  
  
http://localhost:8080/WebGoat
```

✅ 總結：

1. 用 `docker ps` 確認有沒有 port mapping。
2. 確認主機映射的 port（通常是 8080）。
3. 在瀏覽器打 `http://localhost:8080/WebGoat` 進入 WebGoat。

開啟 docker webgoat windows mapping

-2025 windows11 新筆電

我執行狀況 (): 沒開啟

```
系統管理員: 命令提示字元
Microsoft Windows [版本 10.0.26100.4946]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Windows\System32>docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
7730c50d6f27   webgoat/webgoat-8.0:latest          "java -Djava.securit..." 41 minutes ago Up 41 minutes 8080/tcp     flamboyant_galileo

C:\Windows\System32>docker stop webgoat
Error response from daemon: No such container: webgoat

C:\Windows\System32>docker stop flamboyant_galileo
flamboyant_galileo

C:\Windows\System32>docker rm flamboyant_galileo
flamboyant_galileo

C:\Windows\System32>docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0
d34f222b41fd001e1e5401a6e0e5fae711bdf065c53bfa0f8829472b248bf3d0

C:\Windows\System32>docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
d34f222b41fd   webgoat/webgoat-8.0                "java -Djava.securit..." 23 seconds ago Up 22 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
webgoat
```

開啟 docker webgoat windows mapping

-2025 舊筆電

我執行狀況 (): 沒開啟



```
C:\Windows\System32>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e0a38ffabee9	webgoat/webgoat-8.0:latest	"java -Djava.securit..."	6 hours ago	Up 6 hours	8080/tcp	competent_engelbart

我重複開啟兩次, 第二次通知已重疊

```
C:\Windows\System32>docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0
f0cfda09bf4e24cfc898139f9aa3bf788b213c5c1dd81eb64700e29bfa07e9f5
```

```
C:\Windows\System32>docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0
docker: Error response from daemon: Conflict. The container name "/webgoat" is already in use by container "f0cfda09bf4e24cfc898139f9aa3bf788b213c5c1dd81eb64700e29bfa07e9f5". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
```


開啟 docker webgoat windows mapping

-2025

詢問 Chat 如何 remove

when I run "docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0", the system responds with " The container name "/webgoat" is already in use by container "f0cfda09bf4e24cfc898139f9aa3bf788b213c5c1dd81eb64700e29bfa07e9f5". You have to remove (or rename) that container to be able to reuse that name." How Can I remove the container



1. List containers

```
powershell
```

```
docker ps -a
```

You'll see something like:

```
bash
```

CONTAINER ID	IMAGE	STATUS	PORTS	NAMES
f0cfda09bf4e	webgoat/webgoat-8.0	Up 5 minutes	8080/tcp	webgoat

2. Stop the container (if it's running)

```
powershell
```

```
docker stop webgoat
```

開啟 docker webgoat windows mapping

-2025

詢問 Chat 如何 remove

3. Remove the container

```
powershell
```

```
docker rm webgoat
```

4. (Optional) Remove the image too (if you want a clean re-pull)

```
powershell
```

```
docker rmi webgoat/webgoat-8.0
```

5. Run it again with port mapping

```
powershell
```

```
docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0
```

Then access WebGoat at:

```
arduino
```

```
http://localhost:8080/WebGoat
```

開啟 docker webgoat windows mapping

-2025舊筆電

我的執行: **docker ps -a**

```
C:\Windows\System32>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f0cfda09bf4e	webgoat/webgoat-8.0	"java -Djava.securit..."	4 minutes ago	Up 4 minutes	0.0.0.0:8080->8080/tcp	webgoat
e0a38ffabee9	webgoat/webgoat-8.0:latest	"java -Djava.securit..."	7 hours ago	Up 7 hours	8080/tcp	competent_engelbart

docker stop webgoat

```
C:\Windows\System32>docker stop webgoat
```

```
C:\Windows\System32>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f0cfda09bf4e	webgoat/webgoat-8.0	"java -Djava.securit..."	6 minutes ago	Exited (143) 10 seconds ago		webgoat
e0a38ffabee9	webgoat/webgoat-8.0:latest	"java -Djava.securit..."	7 hours ago	Up 7 hours	8080/tcp	competent_engelbart

docker rm webgoat

```
C:\Windows\System32>docker rm webgoat
```

```
C:\Windows\System32>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e0a38ffabee9	webgoat/webgoat-8.0:latest	"java -Djava.securit..."	7 hours ago	Up 7 hours	8080/tcp	competent_engelbart

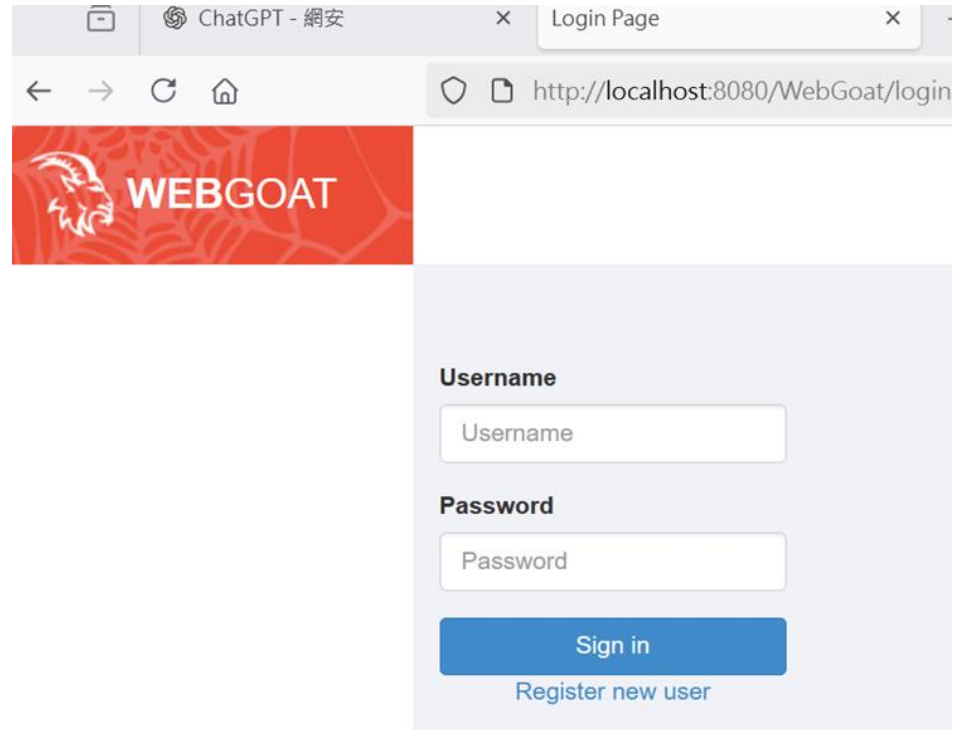
docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0

```
C:\Windows\System32>docker run -d -p 8080:8080 --name webgoat webgoat/webgoat-8.0
c6ae2342bb729a5efd3b48d9729182ec9726de138d1bf6f6eb795d41e813dd49
```

開啟 docker webgoat windows mapping

-2025舊筆電

Docker webgoat mapping 開啟後
<http://localhost:8080/WebGoat>



zenmap

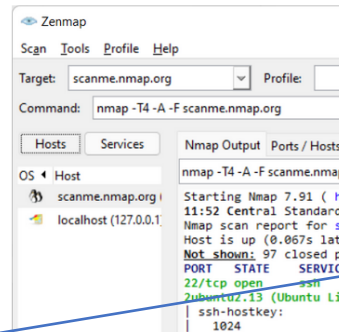
- 改下載 windows 版 zenmap
- <https://nmap.org/download.html>



Get the latest Nmap for your system:

- [Windows](#)
- [macOS](#)
- [Linux \(RPM\)](#)
- [Any other OS \(source code\)](#)

Microsoft Windows binaries



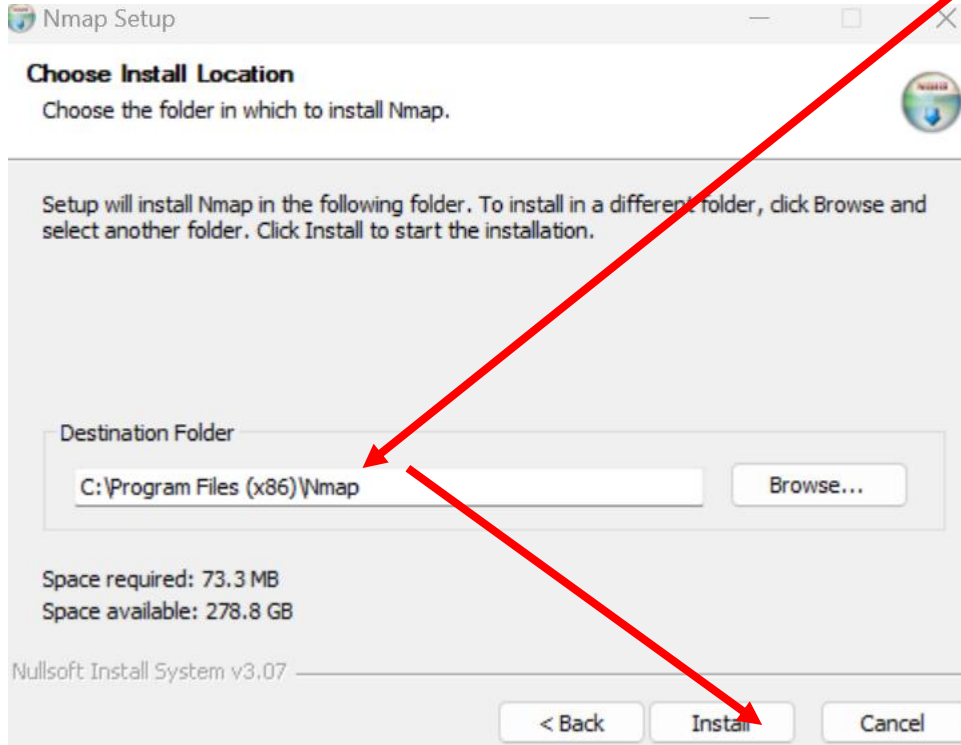
Please read the [Windows section](#) of the Install Guide for limitations and installation instructions for the Windows version of Nmap. It's provided as an executable self-installer which includes Nmap and the Zenmap GUI. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 and newer. We also maintain a [guide for users who must run Nmap on earlier Windows releases](#).

Latest **stable** release self-installer: [nmap-7.94-setup.exe](#)

We have written [post-install usage instructions](#). Please [notify us](#) if you encounter any problems or have suggestions for the installer.

zenmap

- 改下載 windows 版 zenmap
- <https://nmap.org/download.html>



Windows 使用 Zenmap intense scan, all TCP port

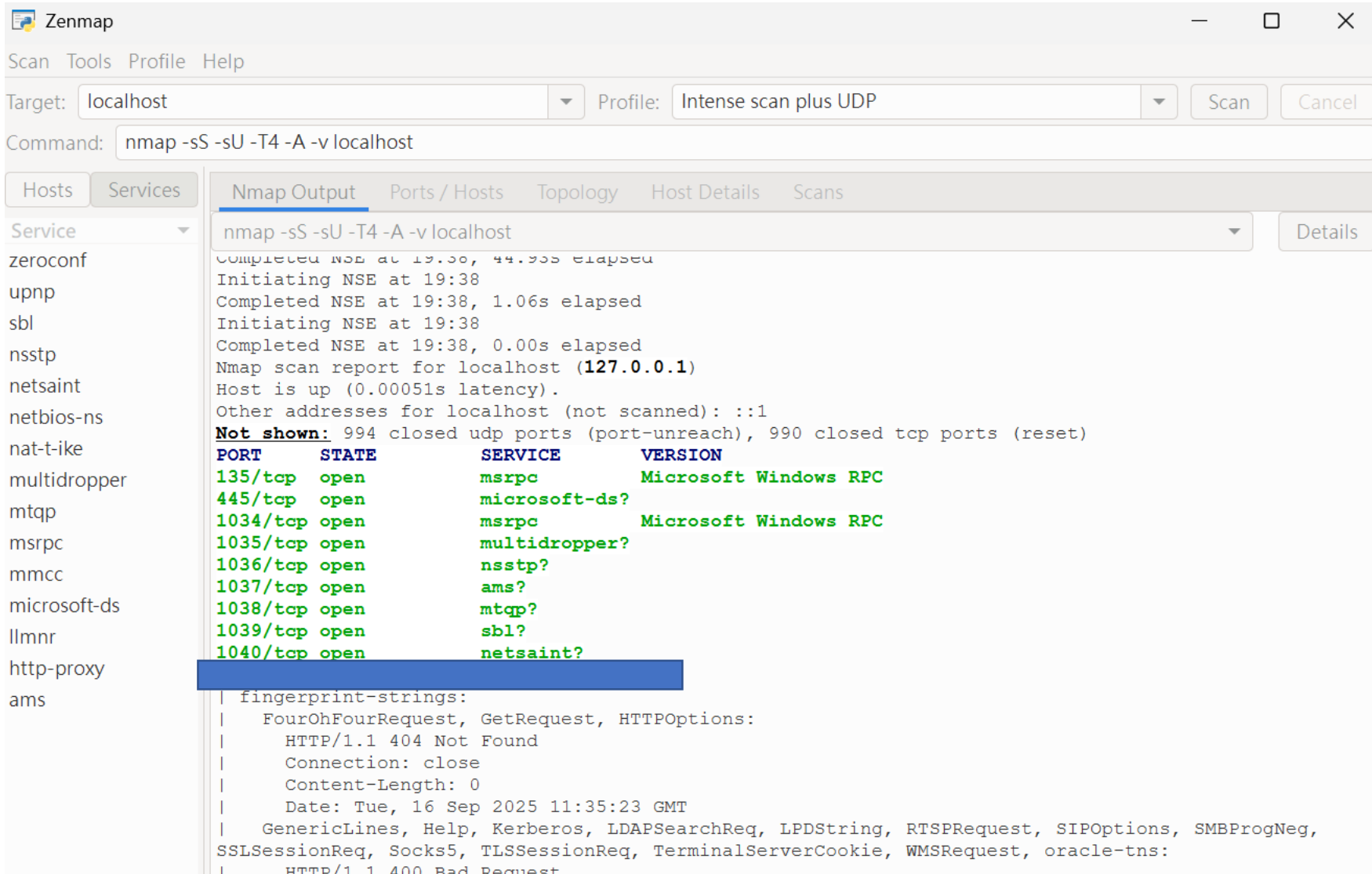
-2025 windows 11 家用版

Before Docker webgoat mapping

Windows 使用 Zenmap intense scan, all TCP port

-2025 windows 11 家用版

After Docker webgoat mapping



Zenmap

Scan Tools Profile Help

Target: localhost Profile: Intense scan plus UDP Scan Cancel

Command: nmap -sS -sU -T4 -A -v localhost

Hosts Services Nmap Output Ports / Hosts Topology Host Details Scans

Service

zeroconf

upnp

sbl

nsstp

netsaint

netbios-ns

nat-t-ike

multidropper

mtqp

msrpc

mmcc

microsoft-ds

llmnr

http-proxy

ams

nmap -sS -sU -T4 -A -v localhost

Completed NSE at 19:38, 44.93s elapsed

Initiating NSE at 19:38

Completed NSE at 19:38, 1.06s elapsed

Initiating NSE at 19:38

Completed NSE at 19:38, 0.00s elapsed

Nmap scan report for localhost (127.0.0.1)

Host is up (0.00051s latency).

Other addresses for localhost (not scanned): ::1

Not shown: 994 closed udp ports (port-unreach), 990 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
445/tcp	open	microsoft-ds?	
1034/tcp	open	msrpc	Microsoft Windows RPC
1035/tcp	open	multidropper?	
1036/tcp	open	nsstp?	
1037/tcp	open	ams?	
1038/tcp	open	mtqp?	
1039/tcp	open	sbl?	
1040/tcp	open	netsaint?	

fingerprint-strings:

FourOhFourRequest, GetRequest, HTTPOptions:

HTTP/1.1 404 Not Found

Connection: close

Content-Length: 0

Date: Tue, 16 Sep 2025 11:35:23 GMT

GenericLines, Help, Kerberos, LDAPSearchReq, LPDString, RTSPRequest, SIPOptions, SMBProgNeg, SSLSessionReq, Socks5, TLSSessionReq, TerminalServerCookie, WMSRequest, oracle-tns:

HTTP/1.1 400 Bad Request

Windows 使用 Zenmap intense scan, all TCP port

-2025 舊筆電

Before Docker webgoat mapping

Zenmap

Scan Tools Profile Help

Target: 127.0.0.1 Profile: Intense scan, all TCP ports

Command: nmap -p 1-65535 -T4 -A -v 127.0.0.1

Hosts Services

OS Host

localhost (127.0.0.1)

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -p 1-65535 -T4 -A -v 127.0.0.1

Scanning 20 services on localhost (127.0.0.1).

Service scan Timing: About 35.00% done; ETC: 08:21 (0:01:12 remaining)

Completed Service scan at 08:20, 83.71s elapsed (20 services on 1 host)

Initiating OS detection (try #1) against localhost (127.0.0.1)

Retrying OS detection (try #2) against localhost (127.0.0.1)

Retrying OS detection (try #3) against localhost (127.0.0.1)

Retrying OS detection (try #4) against localhost (127.0.0.1)

Retrying OS detection (try #5) against localhost (127.0.0.1)

NSE: Script scanning 127.0.0.1.

Initiating NSE at 08:20

Completed NSE at 08:21, 14.23s elapsed

Initiating NSE at 08:21

Completed NSE at 08:21, 0.35s elapsed

Initiating NSE at 08:21

Completed NSE at 08:21, 0.00s elapsed

Nmap scan report for localhost (127.0.0.1)

Host is up (0.00092s latency).

Not shown: 65514 closed tcp ports (reset)

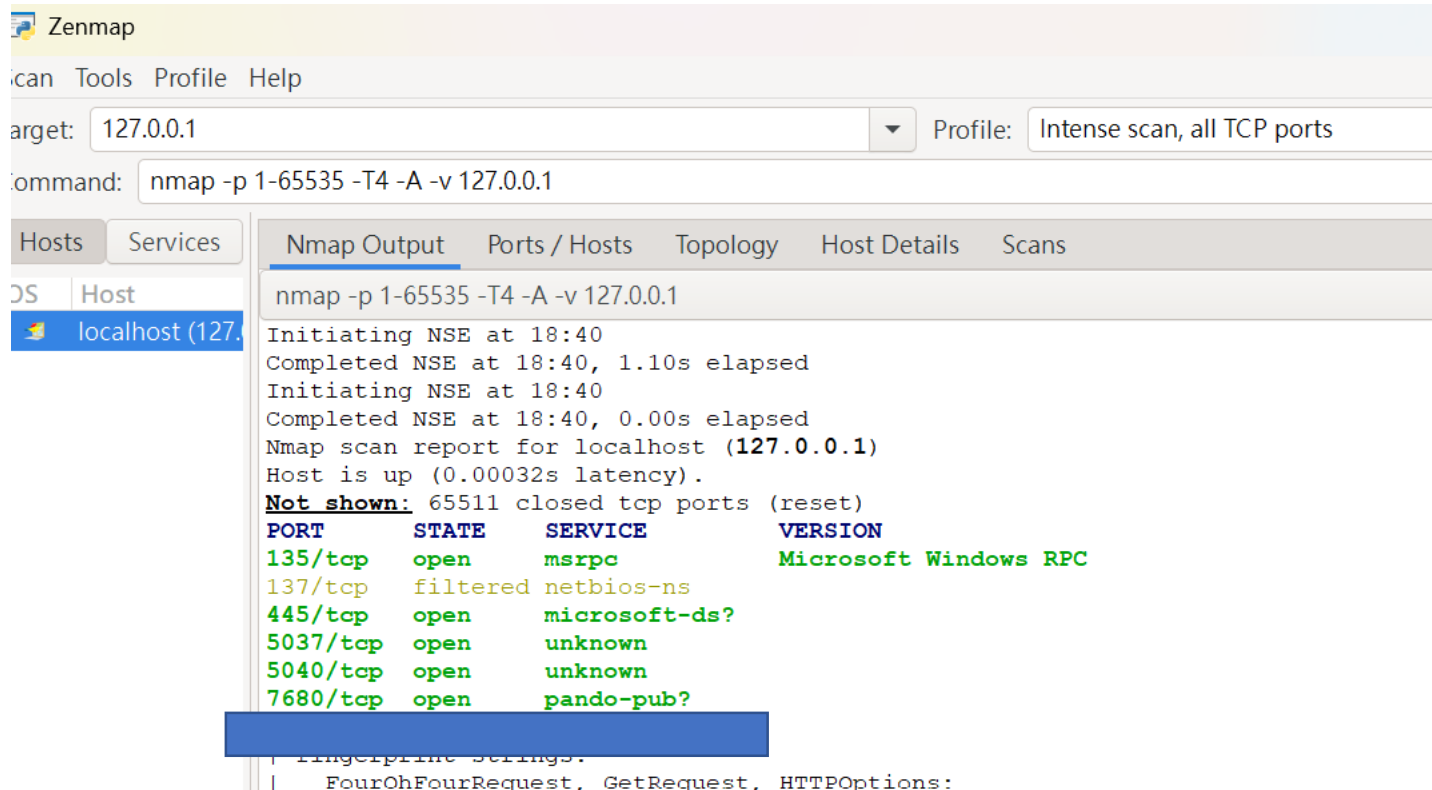
PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
137/tcp	filtered	netbios-ns	
445/tcp	open	microsoft-ds?	
5037/tcp	open	unknown	
5040/tcp	open	unknown	
7680/tcp	open	pando-pub?	
20558/tcp	open	ssl/nagios-nsc	Nagios NSCA

Escalate: Subject: CommonName: 127.0.0.1/organizationName: FIRST COMMERCIAL BANK CO

Windows 使用 Zenmap intense scan, all TCP port

-2025舊筆電

After Docker webgoat mapping



Zenmap

Scan Tools Profile Help

Target: 127.0.0.1 Profile: Intense scan, all TCP ports

Command: nmap -p 1-65535 -T4 -A -v 127.0.0.1

Hosts Services

OS Host

localhost (127.0.0.1)

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -p 1-65535 -T4 -A -v 127.0.0.1

Initiating NSE at 18:40
Completed NSE at 18:40, 1.10s elapsed
Initiating NSE at 18:40
Completed NSE at 18:40, 0.00s elapsed
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00032s latency).
Not shown: 65511 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
137/tcp	filtered	netbios-ns	
445/tcp	open	microsoft-ds?	
5037/tcp	open	unknown	
5040/tcp	open	unknown	
7680/tcp	open	pando-pub?	

Fingerprint strings:
|_ FourOhFourRequest, GetRequest, HTTPOptions:

以下為 2024 windows 環境操作

2024 以 winget install docker.desktop → docker webgoat

以Winget(Windows Package Manager service) upgrade Microsoft.AppInstaller :
winget upgrade Microsoft.AppInstaller



Winget install Docker.DockerDesktop

wsl --install -d Ubuntu

wsl --update

Reboot & initiate dockerDesktop

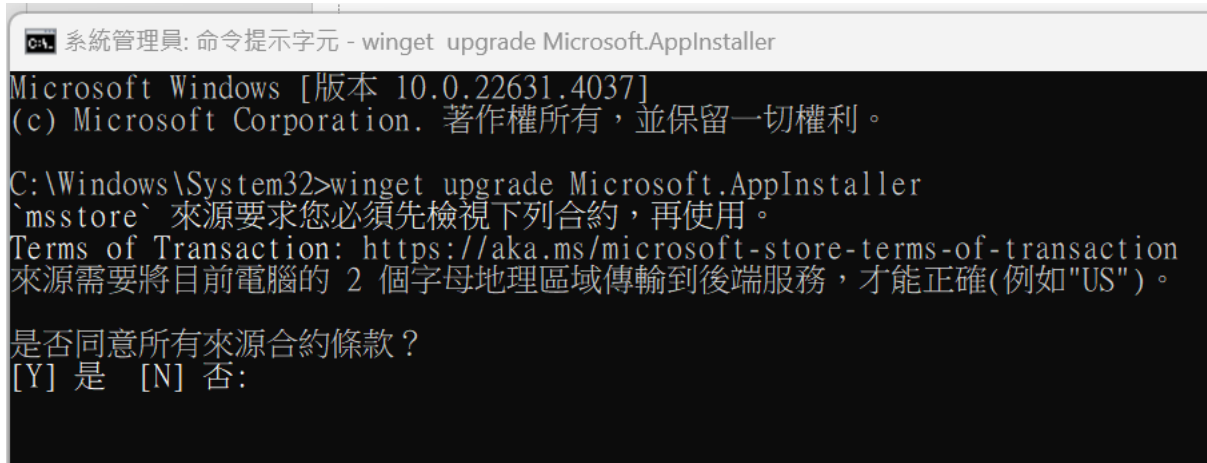
2024 以 winget install docker.desktop ➔ docker webgoat

- Discover, install, upgrade, remove and configure applications on Windows 10 and Windows 11 computers.
- This tool is the client interface to the Windows Package Manager service

[安裝應用程式安裝程式]

[使用 **winget** 工具來更新]

- **winget upgrade Microsoft.AppInstaller**。



```
C:\> 系統管理員: 命令提示字元 - winget upgrade Microsoft.AppInstaller

Microsoft Windows [版本 10.0.22631.4037]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Windows\System32>winget upgrade Microsoft.AppInstaller
`msstore` 來源要求您必須先檢視下列合約，再使用。
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
來源需要將目前電腦的 2 個字母地理區域傳輸到後端服務，才能正確(例如"US")。

是否同意所有來源合約條款？
[Y] 是 [N] 否:
```

2024 以 winget install docker.desktop ➔ docker webgoat

[Use WinGet]

- Winget install docker

```
C:\Windows\System32>winget install docker
找到多個符合輸入條件的套件。請精煉輸入。
名稱                                識別碼                                來源
-----
Dockerun                            9NVBZPBTK78W                        msstore
Docker Desktop Docker.DockerDesktop winget
```

- Winget install Docker.DoclerDesktop

```
C:\Windows\System32>Winget install Docker.DockerDesktop
找到 Docker Desktop [Docker.DockerDesktop] 版本 4.34.2
此應用程式已由其擁有者授權給您。
Microsoft 不負任何責任，也不會授與協力廠商封裝的任何授權。
正在下載 https://desktop.docker.com/win/main/amd64/167172/Docker%20Desktop%20Installer.exe
████████████████████████████████████████████████████████████████████████████████ 490 MB / 490 MB
已成功驗證安裝程式雜湊
正在啟動套件安裝...
已成功安裝
```

➔ 出現在“應用程式”裡 Docker.Desktop ➔ 須以系統管理者執行

2024 以 winget install docker.desktop ➔ docker webgoat

- 出現在“應用程式”裡 Docker.Desktop ➔ 須以系統管理者執行 ➔ 啟動發現 wsl update fail

How is personal diagnostic data handled in Docker Desktop?

When uploading diagnostics to help Docker with investigating issues, the uploaded diagnostics bundle may contain personal data such as usernames and IP addresses. The diagnostics bundles are only accessible to Docker, Inc. employees who are directly involved in diagnosing Docker Desktop issues.

By default, Docker, Inc. will delete uploaded diagnostics bundles after 30 days. You may also request the removal of a diagnostics bundle by either specifying the diagnostics ID or via your GitHub ID (if the diagnostics ID is mentioned in a GitHub issue). Docker, Inc. will only use the data in the diagnostics bundle to investigate specific user issues but may derive high-level (non personal) metrics such as the rate of issues from it.

For more information, see [Docker Data Processing Agreement](#) .

需登出 再重新登入

Docker Desktop - WSL update ➔ failed,
Group membership check

Diagnostic ID: BB75885F-48C4-4427-
9340-FEB84CA34CAE/20240920234550



Group membership check

Your user is a member of the 'docker-users' group, but the changes will only take effect after you log out and back in.

To continue, select Quit and log out and back in.

required compatibility check: Group membership check

Diagnostics ID

BB75885F-48C4-4427-9340-FEB84CA34CAE/20240920234550

Copy diagnostics ID

Quit

2024 以 winget install docker.desktop ➔ docker webgoat

- 出現在 “應用程式” 裡 Docker.Desktop ➔ 須以系統管理者執行 ➔ 啟動發現 wsl update fail

需登出 再重新登入

Docker Desktop - WSL update ➔ failed
wsl --update

Welcome to Docker

Work Personal

Email address
hychien@mail.ncnu.edu.tw

Continue

Or

Create an account

使用 Google 帳戶登入

選擇帳戶

以繼續使用「docker.com」

Docker Desktop - WSL update failed
An error occurred while updating WSL.
You can manually update using `wsl --update` .
If the issue persists, [collect diagnostics and submit an issue](#) .

```
--install -d Ubuntu  
--install --distribution wsl  
Debian  
--inbox
```

[Read our policy for uploaded diagnostic data](#)

Gather diagnostics

2024 以 winget install docker.desktop → docker webgoat

redfish6/

Finish setting up Docker Desktop

version 4.34.2 (167172)

Complete the installation of Docker Desktop.



Use recommended settings (requires administrator password)

Docker Desktop automatically sets the necessary configurations that work for most developers.



Use advanced settings

You manually set your preferred configurations.



Docker Desktop - WSL update failed

An error occurred while updating WSL.

You can manually update using `wsl --update`.

If the issue persists, [collect diagnostics and submit an issue](#).

```
wsl update failed: update failed: updating wsl: ex
1: running WSL command wsl.exe C:\WINDOWS\System32
-update --web-download:
```

```
Copyright (c) Microsoft
Corporation 0Wd\0
```

[Read our policy for uploaded diagnostic data](#)

Gather

Finish

```
C:\Windows\System32>wsl --update
```

Copyright (c) Microsoft Corporation。著作權所有，並保留一切權利。

使用方式: wsl.exe [引數]

引數:

--install <Options>

安裝 Windows 子系統 Linux 版功能。如果未指定選項，建議的功能會隨著預設發佈一起安裝。

若要檢視預設發佈以及其他有效發佈的清單，請使用 'wsl --list --online'。

選項:

--distribution, -d [引數]

指定要下載的發佈，並按名稱安裝。

引數:

有效的發佈名稱（不區分大小寫）。

範例:

```
wsl --install -d Ubuntu
```

2024 以 winget install docker.desktop → docker webgoat

以下列指令解決

- wsl --install -d Ubuntu

```
C:\Windows\System32> wsl --install -d Ubuntu
正在安裝: Windows 子系統 Linux 版
已完成安裝 Windows 子系統 Linux 版。
正在安裝: Ubuntu
已完成安裝 Ubuntu。
已成功執行所要求的操作。請重新開機，變更才能生效。
```

- wsl --update

```
C:\Windows\System32> wsl --update
正在檢查更新。
已安裝最新版的Windows 子系統 Linux 版。
```

- Reboot & initiate docker

 **Docker Desktop - Unexpected WSL error**

An unexpected error occurred while executing a WSL command.

Either shut down WSL down with `wsl --shutdown`, and/or reboot your machine.
You can also try reinstalling WSL and/or Docker Desktop. If the issue persists,
[collect diagnostics and submit an issue](#) .

```
deploying WSL2 distributions
ensuring main distro is deployed: deploying "docker-
desktop": importing WSL distro "無法啟動虛擬網路 - 請執行 :
wsl.exe --install --no-distribution 以安裝選擇性元件 Virtual
Machine Platform\r\nError code:
"
```

[Read our policy for uploaded diagnostic data](#) 

Gather diagnostics

Quit

2024 以 winget install docker.desktop ➔ docker webgoat

以下列指令解決

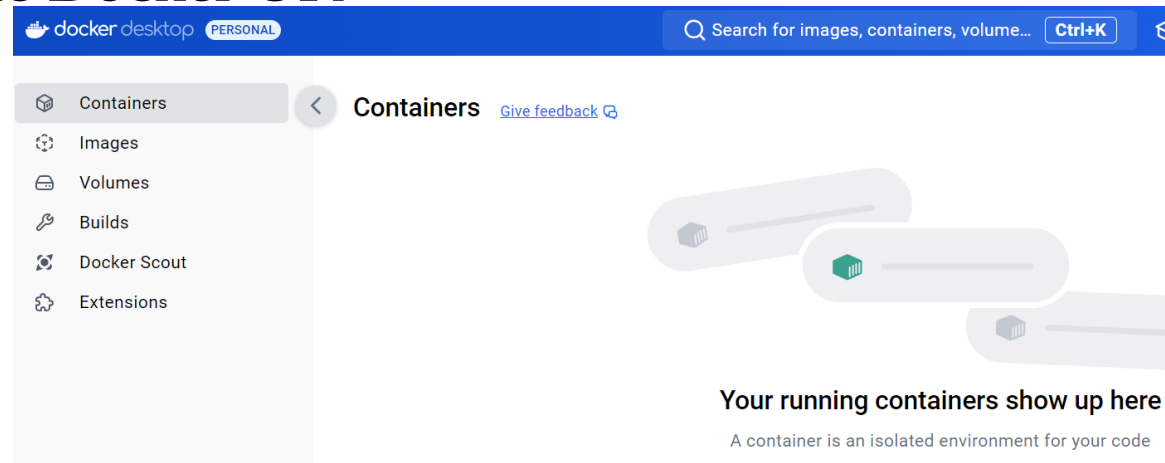
- Winget install wsl

```
C:\Windows\System32>winget install wsl
找到多個符合輸入條件的套件。請精煉輸入。
名稱                                識別碼                                來源
-----
Arch WSL                            9MZNMNKSM73X                          msstore
Windows Subsystem for Linux Microsoft.WSL winget
```

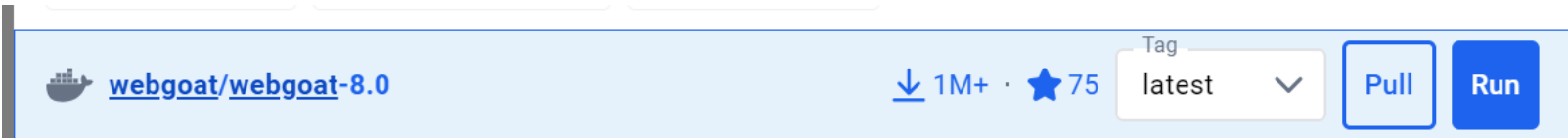
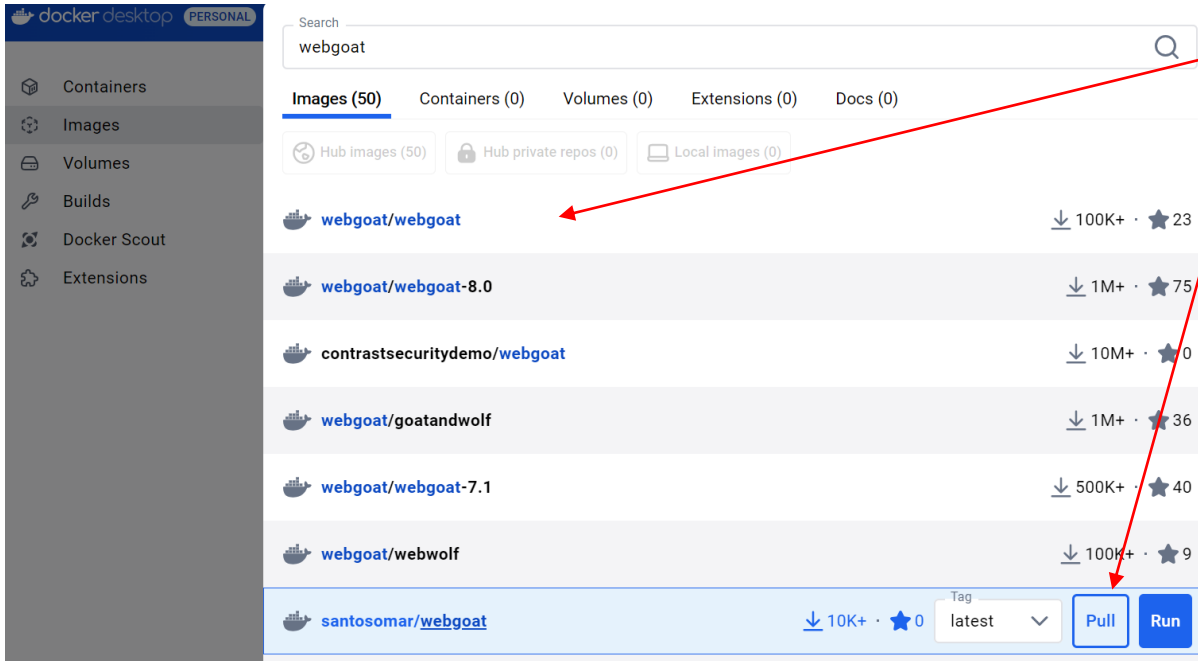
- Winget install Microsoft.WSL

```
C:\Windows\System32>Winget install Microsoft.WSL
找到 Windows Subsystem for Linux [Microsoft.WSL] 版本 2.1.5.0
此應用程式已由其擁有者授權給您。
Microsoft 不負任何責任，也不會授與協力廠商封裝的任何授權。
此套件需要下列相依性：
- Windows 功能
  Microsoft-Windows-Subsystem-Linux
  VirtualMachinePlatform
正在啟用 [Microsoft-Windows-Subsystem-Linux]...
正在啟用 [VirtualMachinePlatform]...
必須重新開機，才能完全啟用 Windows 功能;若要覆寫此檢查，請使用 [--force]。
```

- 重開機 ➔ Initiate Docker OK



2024 Docker webgoat



●



webgoat/webgoat-8.0:latest

^

Webgoat 8.0

Ports

Enter "0" to assign randomly generated host ports.

Host port

```
:8080/tcp
```

Volumes

Host path

Container path

+

Environment variables

Variable

Value

+

Cancel

Run

[Containers](#) / [admiring_beaver](#)

admiring_beaver

 4eeddf06e6c1 

 [webgoat/webgoat-8.0:latest](#)

STATUS

Running (0 seconds ago)



Logs

Inspect

Bind mounts

Exec

Files

Stats

```
2024-09-21 22:49:15 14:49:15.800 [main] INFO org.owasp.webgoat.StartWebGoat - Starting WebGoat with args: --server.port=8080,
--server.address=0.0.0.0
```

2024-09-21 22:49:16

2024-09-21 22:49:16

2024-09-21 22:49:16

2024-09-21 22:49:16

2024-09-21 22:49:16

2024-09-21 22:49:16

2024-09-21 22:49:16

2024-09-21 22:49:16

2024-09-21 22:49:16

```

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  \ \ _ | | | | | | | | | ( _ | ) ) )
  ' | _ | . _ | | _ | | \ , / / / /
=====|_|=====|_|=/ / / / /
:: Spring Boot ::                (v2.2.2.RELEASE)

```

zenmap

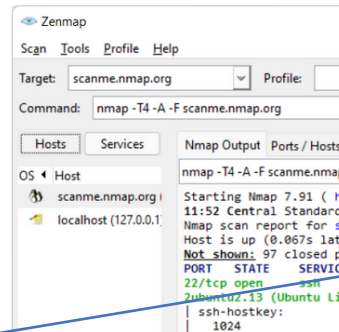
- 改下載 windows 版 zenmap
- <https://nmap.org/download.html>



Get the latest Nmap for your system:

- [Windows](#)
- [macOS](#)
- [Linux \(RPM\)](#)
- [Any other OS \(source code\)](#)

Microsoft Windows binaries



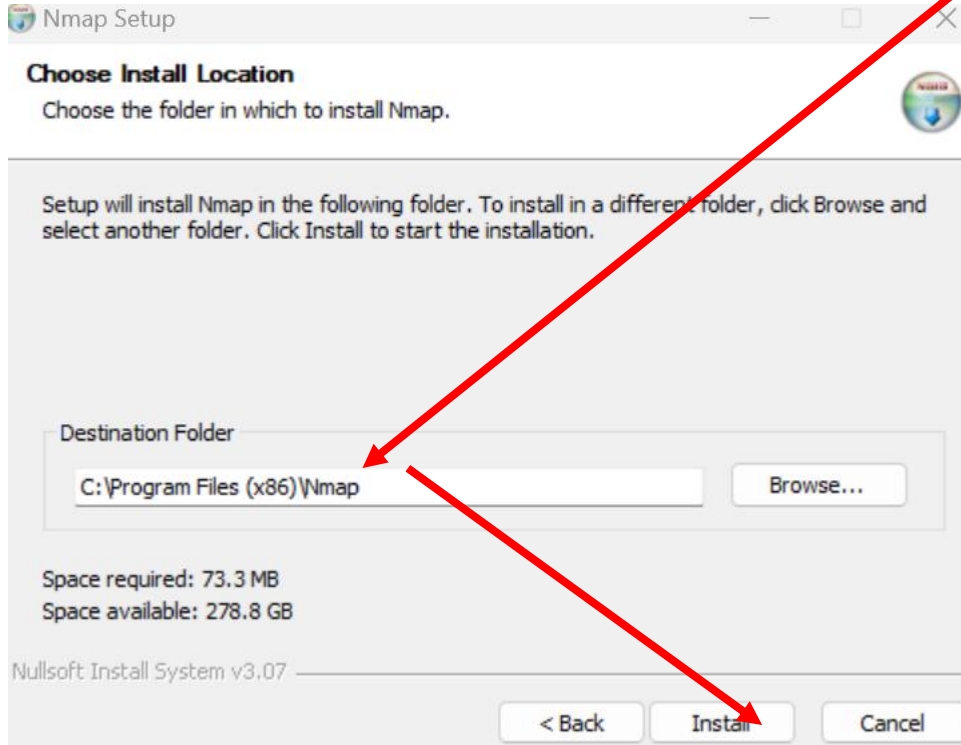
Please read the [Windows section](#) of the Install Guide for limitations and installation instructions for the Windows version of Nmap. It's provided as an executable self-installer which includes Nmap and the Zenmap GUI. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 and newer. We also maintain a [guide for users who must run Nmap on earlier Windows releases](#).

Latest **stable** release self-installer: [nmap-7.94-setup.exe](#)

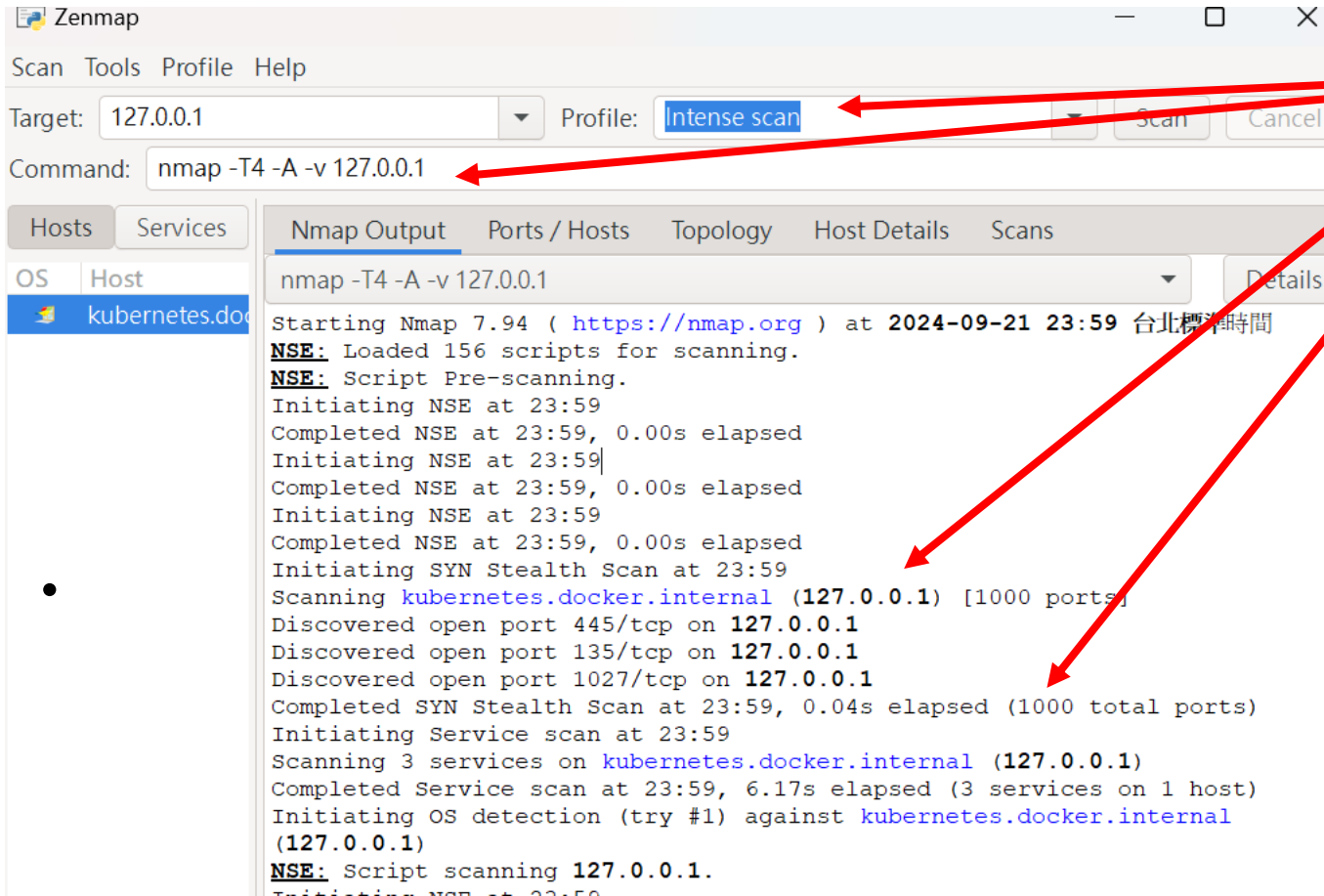
We have written [post-install usage instructions](#). Please [notify us](#) if you encounter any problems or have suggestions for the installer.

zenmap

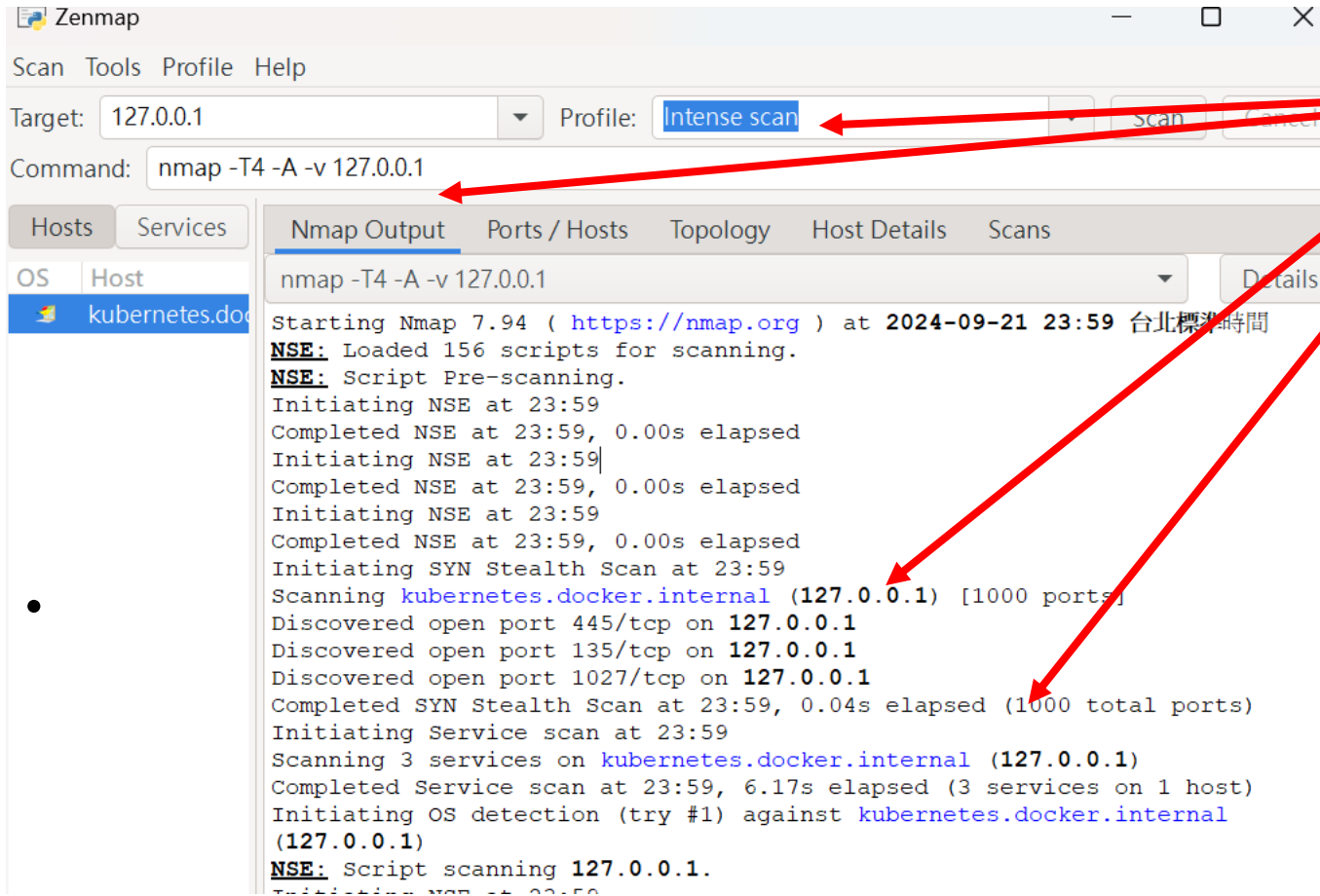
- 改下載 windows 版 zenmap
- <https://nmap.org/download.html>



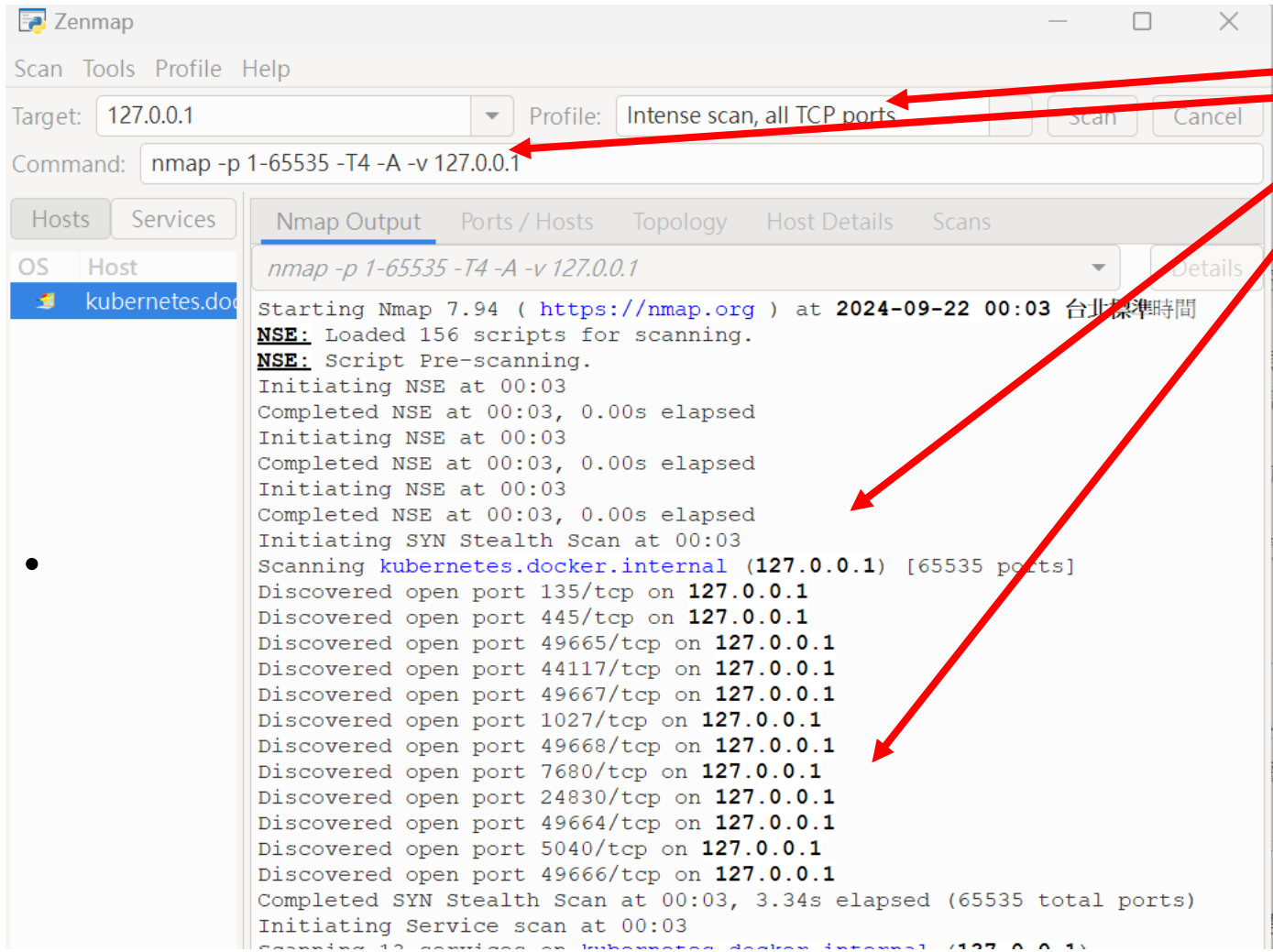
zenmap



zenmap



zenmap



SYN Stealth Scan

A TCP SYN scan is a **stealth scan used to determine if ports on a target system are open, closed or filtered.**

Nmap sends a SYN packet to the target and waits for a response. If the target responds with a SYN/ACK packet, the port is considered open and ready to establish a connection.

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